Candidate's Name: ……

Signature: ……

Random No.						Personal No.		

(*Do not write your School / Center Name or Number anywhere on this booklet.*)

545/1

CHEMISTRY

Paper 1

April,2026

2 hours

For Scorers' Use Only

Total Weighted Score		
Initials		

CHEMISTRY DEPARTMENT EXAMINANTIONS BOARD

Uganda Certificate of Education

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of **two** *sections;* **A** *and* **B**.

It has **six** *examination items.*

Section **A** *has* **two compulsory** *items. Responses to section* **A must** *be written in the spaces provided.*

Section **B** *has* **two** *parts;* **I** *and* **II.** *Respond to only* **one** *item from* **each** *part.*

Responses to section **B must** *be written in the answer sheet*(s) *provided.*

Respond to **four** *items in all.*

Responses to each item in section **B** *should start on a fresh page.*

Any additional item(s) *responded to will* **not** *be scored.*

For Scorers' Use Only

Item No.	Basic Code	Weighted Score	Scorers' Initials
1	T		
	D		
	E		
2	C		
	X		
	I		
_____(3/4)	P		
	S		
	B		
_____(5/6)	N		
	M		
	Be		
Total Weighted Score			

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SECTION A

Respond to **all** *the items in this section in the spaces provided.*

Item 1

Sharif and John are senior 1 students who are part of the school football team that went to play a football match with another school. During the match, they slipped and fallen down several times hence their white shirts became dirty.
Upon going back home, Sharif decided to use a product **Q** to wash his shirt with borehole water and his shirt was cleaned instantly after one rinse. John decided to use another product **P** with borehole water but his shirt remained with brown stains despite rinsing it several times.
John remained confused about his outcome and he would like to compare the product he used to that used by Sharif and he has approached you for help.

Task:

(a) State the categories of the products and their function.

_____

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_____

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_____

For Scorer's Use Only

Basis Code	**T**
Score	

(b) Explain the side effect(s) associated with the products plus mitigation.

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_____

For Scorer's Use Only

Basis Code	**D**
Score	

(c) State the;

(i) similarities of the products.

_____

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_____

_____

(ii) Differences between the products.

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_____

For Scorer's Use Only

Basis Code	**E**
Score	

Item 2

Students of Havilah primary school went for a trip to a jewelry store. Many students were amazed by the brown coloured necklaces and medals hence they reached out to the jewelry store owner who told the them that the necklaces and medals are made from substance **T** which makes the jewelry materials to be of good quality.

The jewelry store owner further cautioned the students about the impact of such materials when disposed to the environment. After coming back home, many students remained confused about the message from the jewelry store owner and they approached you for help. Figure **1** shows necklaces and medals.



Fig. 1. Necklaces and medals.

Task:

(a) Explain the nature of the substance **T**, and identify it.

_____

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For Scorer's Use Only

Basis Code	C
Score	

(b) Explain the properties of the substance **T**.

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(c) Give any other use of the substance **T**.

_____

_____

For Scorer's Use Only

Basis Code	X
Score	

(d) Write **one** impact of having a lot of substance **T** in the environment and how it can be mitigated.

_____

_____

_____

_____

For Scorer's Use Only

Basis Code	I
Score	

SECTION B

Part I

Respond to only **one** *item in this part in the answer booklet*(*s*) *provided.*

Item 3

Following the increased rate of poor quality and dirty water from the boreholes in Kisumu town, many residents in the town were suffering with poor cleaning of their household laundry.
A local investor has set up a soapless soap manufacturing plant in the town. Figure **2** shows a soapless soap manufacturing plant.



Fig. 2. Soapless soap manufacturing plant.

The residents in the town have requested to be guided about the production process of soapless soap, how any side effect(s) is/are will be mitigated plus the social and economic benefits of the manufacturing plant in their town.
A television program has been organized to address the request by the residents and the town mayor has chosen you to present during the television program.

Task:
Write down the key details you will present during the television program.

Item 4

Ethanol is an essential component in sanitizers. Following an increased demand for sanitizers countrywide due to Covid-19 spike, the government has hired an investor to set up an ethanol brewing plant to make more ethanol to be added in sanitizers.
Aware of the merits and demerits of such a plant, your school science club has decided to organize a radio talk show to be aired nationwide, sensitizing the citizens how ethanol with be produced, the social and economic benefits of the brewing plant to the country, and how any side effect(s) is/are to be mitigated.
You have been asked to present during the radio talk show.

Task:
Make a write-up you will present.

Part II

Respond to only **one** *item from this part in the answer booklet*(*s*) *provided.*

Item 5

Following the government's ban on cutting down trees, several residents resorted to buying natural gas cylinders to be used for cooking at their homes.
However, there has been a rampant increase in number of cases showing various dangers experienced by the citizens due to improper use of nature gas.
In order to address this issue, the government has decided to organize a nationwide campaign to sensitize the citizens about the use of the natural gas.
The government has hired you to be the campaign champion to address the citizens.
Figure **3** shows nature gas cylinders.



Fig. 3. Nature gas cylinders

Task:
Make a write-up of the message you will present during the campaign

Item 6

A youth group has a washing bay along the river. As they wash vehicles, the waste water drains into the river.
When senior two students from a nearby school went for a chemistry study tour to the river, they found the youth busy with their routine work. One of the students noticed that the water-body around the washing bay looked abnormal.
The student expressed worry about the safety of the people around the river and drew the attention of the area local council. In response, the local council has called for a meeting with the youth.
You have been selected to give a sensitization talk.

Task:
Write details of the key issues you will address during the sensitization talk.